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Paul Ehrenfest in Memoriam

IT HAPPENS SO often nowadays that men of high qualities depart this life of their own free will that we no longer feel such a conclusion to be unusual. Yet the decision to take leave generally stems from an incapacity—or at least an unwillingness—to resign oneself to new and more difficult *outward* conditions of life. To refuse to live out one's natural life because of *inner* conflicts that are felt to be intolerable—that is even today in persons of sound mind a rare occurrence, possible only in the case of the noblest and morally most exalted personalities. It is to such a tragic inner conflict that our friend Paul Ehrenfest has succumbed. Those who knew him well, as was vouchsafed to me, know that this unblemished personality in the main fell victim to a conflict of conscience that in some form or other is spared no university teacher who has passed, say, his fiftieth year.

I came to know him twenty-two years ago. He visited me in Prague, coming straight from Russia where he as a Jew was debarred from teaching at institutions of higher learning. He was looking for a sphere of work in central or western Europe. But we talked little of that, for it was the state of science at the time that took up almost all of our interest. Both of us realized that classical mechanics and the theory of the electric field had failed in the face of the phenomena of heat radiation and molecular processes (the statistical theory of heat), but there seemed to be no feasible way out of this dilemma. The logical gap in Planck's Theory of Radiation—which we both, nevertheless, greatly admired—was apparent to us. We also discussed the Theory of Relativity, to which he responded with a certain skepticism but with the critical judgment peculiar to him. Within a few hours we were true friends—as though our dreams and aspirations were meant for each other. We remained joined in close friendship until he departed this life.

His stature lay in his unusually well developed faculty to grasp the essence of a theoretical notion, to strip a theory of its mathematical accouterments until the simple basic idea emerged with clarity. This capacity made him a peerless teacher. It was on its account that he was invited to scientific congresses; for he always brought clarity and acuteness into any discussion. He fought against fuzziness and circumlocution, when necessary employing his sharp wit and even downright discourtesy. Some of his utterances could have been interpreted almost as arrogant, yet his tragedy lay precisely in an almost morbid lack of self-confidence. He suffered incessantly from the fact that his critical faculties transcended his constructive capacities. In a manner of speaking, his critical sense robbed him of his love for the offspring of his own mind even before they were born.

Shortly after our first encounter there occurred the great turning-point in Ehrenfest's outward career. Our revered master, Lorentz, anxious to retire from regular university teaching, had recognized Ehrenfest for the inspired teacher that he was and recommended him as his successor. A marvelous sphere of activity opened up to the still youthful man. He was not merely the best teacher in our profession whom I have ever known; he was also passionately preoccupied with the development and destiny of men, especially his students. To understand others, to gain their friendship and trust, to aid anyone embroiled in outer or inner struggles, to encourage youthful talent—all this was his real element, almost more than immersion in scientific problems. His students and colleagues in Leyden loved and esteemed him. They knew his utter devotion, his nature so wholly attuned to service and help. Should he not have been a happy man?

In truth he felt unhappier than anyone else who was close to me. The reason was that he did not feel equal to the lofty task that confronted him. Of what use was it that everyone held him in esteem? His sense of inadequacy, objectively unjustified, plagued him incessantly, often robbing him of the peace of mind necessary for tranquil research. So greatly did he suffer that he was compelled to seek solace in distraction. His frequent aimless travels, his preoccupation with the radio, and many other features of his restless life stemmed not from a need for composure and harmless hobbies but rather from a curious urge for escape caused by the psychic conflict at which I have hinted.

In the last few years this situation was aggravated by the strangely turbulent development which theoretical physics has recently undergone. To learn and to teach things that one cannot fully accept in